



Nanyang Junior College
Chemistry (9476)

Experiment 7
Thermochemistry

H2

JC1/2026

Name: Suggested Answers

CT: 26 Tutor:

Marks

/ 25

Learning Objectives

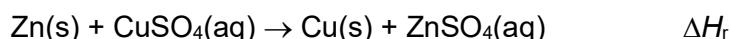
- To conduct a thermochemistry experiment involving temperature measurements
- To measure temperature at regular time intervals for graphical analysis
- To perform a calibration to determine the heat capacity of the polystyrene cup (calculations only)

Determination of the enthalpy change of reaction, ΔH_r , between zinc and copper(II) sulfate

NY 1 is zinc powder.

NY 2 is 0.500 mol dm⁻³ copper(II) sulfate solution.

The enthalpy change of reaction, ΔH_r , is the energy change when molar quantities of reactants as stated in the thermochemical equation react together under standard conditions. In this experiment, you will determine the enthalpy change for the following reaction.



You will obtain the temperature change for this reaction using two different methods.

(I) Direct measurement

Upon mixing the reactants, you will record the maximum (or minimum) temperature reached to obtain the temperature change for the reaction.

(II) Graphical method

You will measure the temperature of the contents in a polystyrene cup at regular time intervals, before and after the zinc powder is added. You will then analyse your results graphically to obtain a value for the temperature change caused by the reaction.

You will use the value for the temperature change to calculate the heat change, q , for the experiment.

$$q = m_{\text{solution}} \times c_{\text{solution}} \times \Delta T_{\text{solution}}$$

The heat change is then used to determine a value for the enthalpy change of the reaction, ΔH_r .

$$\Delta H_r = \pm \frac{\text{heat evolved or absorbed, } q}{\text{amount of substance, } n}$$

(a) Direct measurement of temperature change

1. Weigh accurately about 1.5 g of **NY 1** in a weighing bottle using an electronic mass balance. Record the mass in the space provided below.
2. Place a polystyrene cup inside a 250 cm³ beaker.
3. Use a 50 cm³ measuring cylinder to transfer 30.0 cm³ of **NY 2** into the polystyrene cup.
4. Carefully stir the **NY 2** in the polystyrene cup with a 0.2 °C division thermometer. Read and record the initial temperature, T_i .
5. Rapidly tip all the **NY 1** from the weighing bottle into the polystyrene cup. Stir the mixture.
6. Record the highest temperature reached, $T_{\max}$, and determine the maximum temperature rise, ΔT , for the reaction.
7. Reweigh the weighing bottle and record this mass.

Table 1 – Mass of NY 1

Mass of weighing bottle and NY 1 / g	
Mass of empty weighing bottle / g	
Mass of weighing bottle and residue / g	
Mass of NY 1 used / g	1.512

Table 2 – Temperature

Highest temperature reached, $T_{\max}$ / °C	46.7
Initial temperature, T_i / °C	30.0
Temperature rise, ΔT / °C	16.7

For Table 2,

[1] correct headers and units

[1] record to 1 d.p.

[2]

(b) Measurement of temperature at regular time intervals for graphical analysis

1. Weigh accurately about 1.5 g of **NY 1** in a weighing bottle using an electronic mass balance. Record the mass in the space provided below.
2. Place a polystyrene cup inside a 250 cm³ beaker.
3. Use a 50 cm³ measuring cylinder to transfer 30.0 cm³ of **NY 2** into the polystyrene cup.
4. Carefully stir the **NY 2** in the polystyrene cup with a 0.2 °C division thermometer. Read and record its temperature, T (time, $t = 0.0$ min) in the space provided on the next page.
5. Continue to stir **NY 2**. Read and record T every 0.5 minute for two minutes.
6. At **exactly** 2.5 minutes, transfer all the **NY 1** from the weighing bottle into the polystyrene cup. Stir the mixture but do not read T .
7. Continue to stir the mixture. Read and record T at $t = 3.0$ min and every 0.5 min until $t = 9.0$ min.
8. Reweigh the weighing bottle and record this mass.

Table 3 – Mass of **NY 1**

Mass of weighing bottle and NY 1 / g	
Mass of empty weighing bottle / g	
Mass of weighing bottle and residue / g	
Mass of NY 1 used / g	1.500

For Table 1 and Table 3,

[1] correct headers and units

[1] record to 3 d.p. and $1.450 \text{ g} \leq \text{mass of NY 1} \leq 1.549 \text{ g}$

Table 4 – Temperature

t / min	$T / ^\circ\text{C}$	t / min	$T / ^\circ\text{C}$
0.0	29.6	5.0	45.9
0.5	29.6	5.5	45.8
1.0	29.6	6.0	45.5
1.5	29.6	6.5	45.0
2.0	29.6	7.0	45.0
2.5	—	7.5	44.7
3.0	42.0	8.0	44.4
3.5	45.0	8.5	44.3
4.0	45.8	9.0	43.8
4.5	45.8		

For Table 4,

[1] correct headers and units

[1] record to 1 d.p. for T

[1] no T reading at $t = 2.5$ min; reads T every 0.5 min until $t = 9.0$ min

[5]

- (c) Using your results in (b), plot a graph of temperature, T , on the y-axis, against time, t , on the x-axis on the grid provided on the next page. **Your scale on the y-axis should allow for extrapolation above the highest temperature recorded.**

Draw a best-fit straight line taking into account all of the points before $t = 2.5$ min.

Draw another best-fit straight line taking into account all of the points after the temperature of the mixture has started to fall steadily.

Extrapolate (extend) both lines to $t = 2.5$ min.

[1] label axes with units

[1] appropriate choice of scale + all points plotted

[1] two best fit straight lines + extrapolate to $t = 2.5$ min

[3]

- (d) From your graph, read the minimum temperature, $T_{\min}$, and the maximum temperature, $T_{\max}$, at $t = 2.5$ min. Record these values in the spaces below.

Calculate the temperature change, ΔT , at $t = 2.5$ min.

$$T_{\min} = \underline{\quad 29.6 \quad} ^\circ\text{C}$$

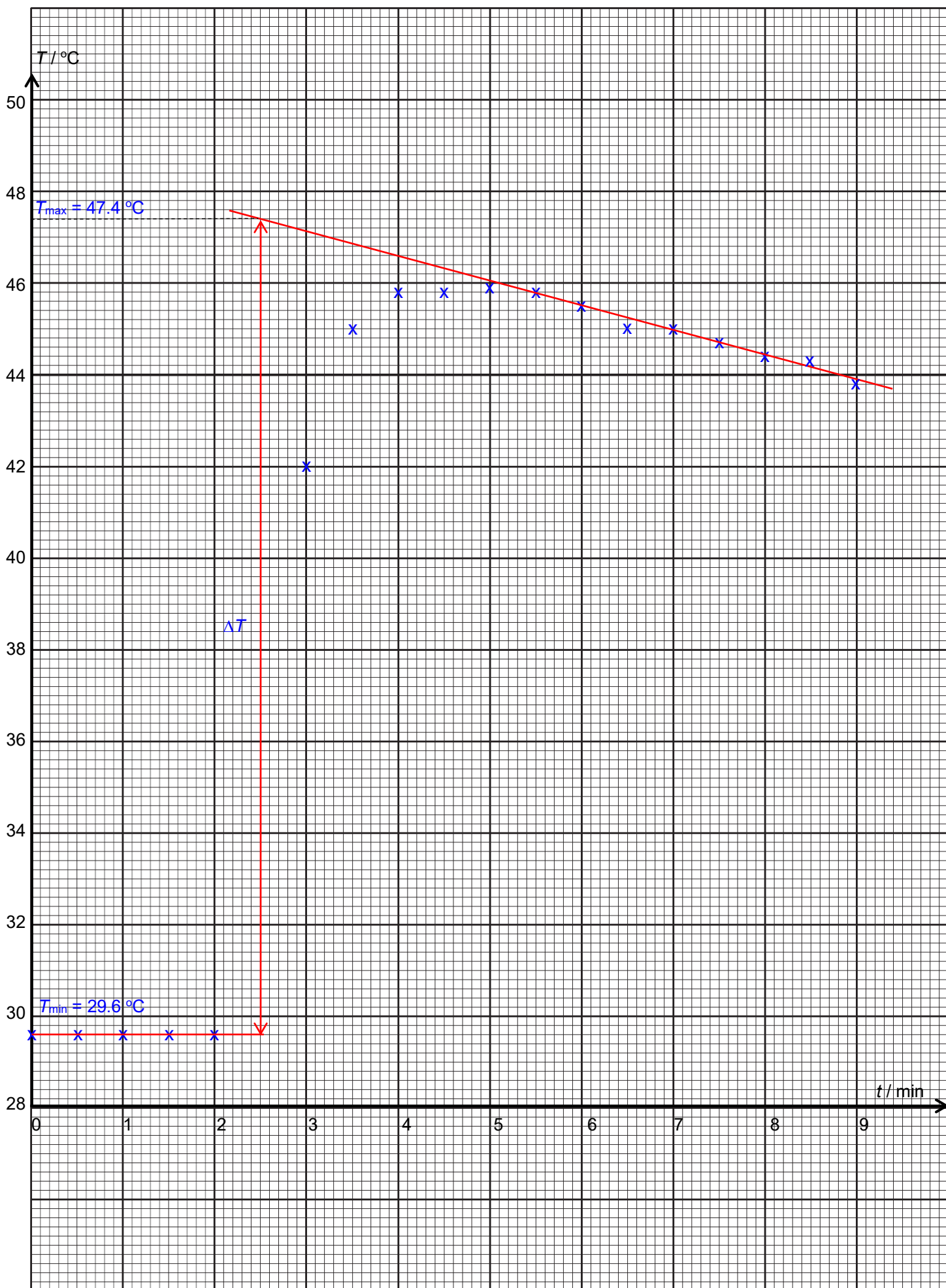
$$\Delta T = 47.4 - 29.6 = 17.8 ^\circ\text{C}$$

$$T_{\max} = \underline{\quad 47.4 \quad} ^\circ\text{C}$$

$$\Delta T = \underline{\quad 17.8 \quad} ^\circ\text{C}$$

[1] $T_{\min}$ and $T_{\max}$ read correctly + ΔT calculated correctly

[1]



Calculations

- (e) Calculate the heat released, q , for your experiment using the value you deduced in (d).

Assume that the specific heat capacity of the reaction mixture is $4.18 \text{ J g}^{-1} \text{ K}^{-1}$, and that the density of the reaction mixture is 1.00 g cm^{-3} .

$$\text{Heat released, } q = 30.0 \times 1.00 \times 4.18 \times 17.8 = 2232 = 2230 \text{ J [1]}$$

$$q = \underline{\quad 2230 \text{ J} \quad} [1]$$

- (f) Calculate the enthalpy change of reaction, ΔH_r , between Zn(s) and $\text{CuSO}_4(\text{aq})$.
[A_r : Zn, 65.4]

$$n_{\text{Zn}} = \frac{1.500}{65.4} = 0.02293 \text{ mol}$$

$$n_{\text{CuSO}_4} = \frac{30.0}{1000} \times 0.500 = 0.01500 \text{ mol}$$

$\Rightarrow \text{CuSO}_4$ is the limiting reactant. [1]

$$\Delta H_r = -\frac{2232}{0.01500} \times 10^{-3} = -148.8 = -149 \text{ kJ mol}^{-1} [1]$$

[1] negative sign and units

$$\Delta H_r = \underline{\quad -149 \text{ kJ mol}^{-1} \quad} [3]$$

Analysis and Evaluation

- (g) Calculate, to 2 s.f., the percentage error in measuring a temperature change of 5.0°C using a 0.2°C division thermometer

$$\text{Percentage error} = \pm \frac{2 \times 0.1}{5.0} \times 100 = \pm 4.0 \% [1]$$

$$\text{percentage error in measuring temperature change} = \underline{\quad \pm 4.0 \quad} \% [1]$$

- (h) Suggest which method, direct measurement method in (a) or graphical method in (b), gives a more accurate value for temperature change, ΔT . Explain your answer.

The graphical method gives a more accurate value than the direct measurement method. [1]

This is because the graphical method accounts for the heat lost to surroundings. [1]

[2]

When an *exothermic* reaction takes place in a container such as a polystyrene cup, some of the evolved heat energy is absorbed by the polystyrene cup. When an *endothermic* reaction takes place, some of the required heat energy is supplied by the polystyrene cup.

Samples of hot water can be added to cold water in the polystyrene cup to determine its approximate heat capacity so as to account for this heat energy difference.

The amount of heat energy evolved or supplied for a 1 °C change in temperature is known as the heat capacity of the cup. Note the difference between heat capacity, C and specific heat capacity, c .

Heat capacity of a substance, C = mass of substance, m × specific heat capacity, c

(i) **Determination of the heat capacity of the polystyrene cup**
(DO NOT CARRY OUT THIS EXPERIMENT)

1. Place a polystyrene cup inside a 250 cm³ beaker.
2. Use a 100 cm³ measuring cylinder to transfer 50.0 cm³ of tap water into the polystyrene cup.
3. Measure the temperature of the tap water in the polystyrene cup with a 0.2 °C division thermometer and record the temperature, T_1 .
4. Use the 100 cm³ measuring cylinder to transfer 50.0 cm³ of tap water into a 100 cm³ beaker. Heat the beaker with a Bunsen burner, with stirring, until the temperature of the water has increased by about 20 °C.
5. Stir the hot water in the beaker. Measure and record the temperature, T_2 .
6. Immediately add the hot water from the beaker into the polystyrene cup and stir. **Allow the temperature to stabilise.** Measure and record the temperature of the combined water, T_f .

Results

$T_1 / ^\circ\text{C}$	28.7
$T_2 / ^\circ\text{C}$	60.5
$T_f / ^\circ\text{C}$	43.2

Temperature rise for the tap water in the polystyrene cup = 43.2 – 28.7 = 14.5 °C

Temperature fall for the hot water in the 100 cm³ beaker = 60.5 – 43.2 = 17.3 °C

[1]

[1] for both

Calculations

You may assume that the specific heat capacity of water is $4.18 \text{ J cm}^{-3} \text{ K}^{-1}$.

- (j) Calculate the heat energy gained by the 50.0 cm^3 tap water in the polystyrene cup.

$$\text{Heat gained} = 50.0 \times 4.18 \times 14.5 = 3030.5 = 3030 \text{ J [1]}$$

$$\text{heat energy gained by the tap water} = \underline{\quad 3030 \quad} \text{ J [1]}$$

- (k) Calculate the heat energy lost by the 50.0 cm^3 hot water from the beaker.

$$\text{Heat lost} = 50.0 \times 4.18 \times 17.3 = 3615.7 = 3620 \text{ J [1]}$$

$$\text{heat energy lost by the hot water} = \underline{\quad 3620 \quad} \text{ J [1]}$$

The difference between the values calculated in (j) and (k) is an approximate value for the heat energy absorbed by the polystyrene cup during the experiment.

The heat capacity of the polystyrene cup is the amount of heat energy absorbed for a 1°C change in temperature.

$q = C_{\text{polystyrene cup}} \times \Delta T$
--

- (l) Use your answers in (i), (j) and (k) to calculate the heat capacity of the polystyrene cup.

$$\text{Heat lost by hot water} = \text{Heat gained by tap water} + \text{Heat gained by polystyrene cup}$$

$$3615.7 = 3030.5 + \text{heat gained by polystyrene cup [1]}$$

$$\text{Heat capacity of polystyrene cup} = \frac{3615.7 - 3030.5}{14.5} = 40.35 = 40.4 \text{ J } ^\circ\text{C}^{-1} \text{ [1]}$$

$$\text{heat capacity of polystyrene cup} = \underline{\quad 40.4 \quad} \text{ J } ^\circ\text{C}^{-1} \text{ [2]}$$

- (m) Use the heat capacity of the polystyrene cup determined in (l) and your answer in (e) to calculate another value for the enthalpy change of reaction, ΔH_r , between Zn(s) and $\text{CuSO}_4(\text{aq})$.

$$\begin{aligned} \text{Heat released, } q &= \text{Heat gained by solution} + \text{heat gained by polystyrene cup} \\ &= 2232 + (40.35 \times 17.8) \\ &= 2950 \text{ J [1]} \end{aligned}$$

$$\Delta H_r = -\frac{2950}{0.01500} \times 10^{-3} = -196.6 = -197 \text{ kJ mol}^{-1} \text{ [1] with negative sign}$$

$$\Delta H_r = \underline{\quad -197 \quad} \text{ kJ mol}^{-1} \text{ [2]}$$